

Database Systems Lab

Lab Task 6



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24P-0554

Section 4C

Operation 01 & 02

```
CREATE DATABASE 24P_0554;  
SHOW DATABASES;
```

```
MariaDB [(none)]> CREATE DATABASE 24P_0554;  
ERROR 1007 (HY000): Can't create database '24p_0554'; database exists  
MariaDB [(none)]> SHOW DATABASES;  
+-----+  
| Database |  
+-----+  
| 24p_0554 |  
| information_schema |  
| mysql |  
| performance_schema |  
| phpmyadmin |  
| test |  
+-----+  
6 rows in set (0.002 sec)  
  
MariaDB [(none)]>
```

Operation 03

```
USE 24P_0554;
```

```
MariaDB [(none)]> USE 24P_0554;  
Database changed  
MariaDB [24P_0554]> |
```

Operation 04

```

8 CREATE TABLE publishers (
9     publisher_id INT NOT NULL AUTO_INCREMENT,
10     name VARCHAR(255) NOT NULL,
11     founded_year YEAR,
12     website VARCHAR(255) UNIQUE,
13     annual_revenue DECIMAL(15,2),
14     PRIMARY KEY (publisher_id)
15 ); 34ms
16

```

```
CREATE TABLE publishers (
publisher_id INT NOT NULL AUTO_INCREMENT,
    name VARCHAR(255) NOT NULL,
    founded_year YEAR,
    website VARCHAR(255) UNIQUE,
    annual_revenue DECIMAL(15,2),
    PRIMARY KEY (publisher_id)
);
```

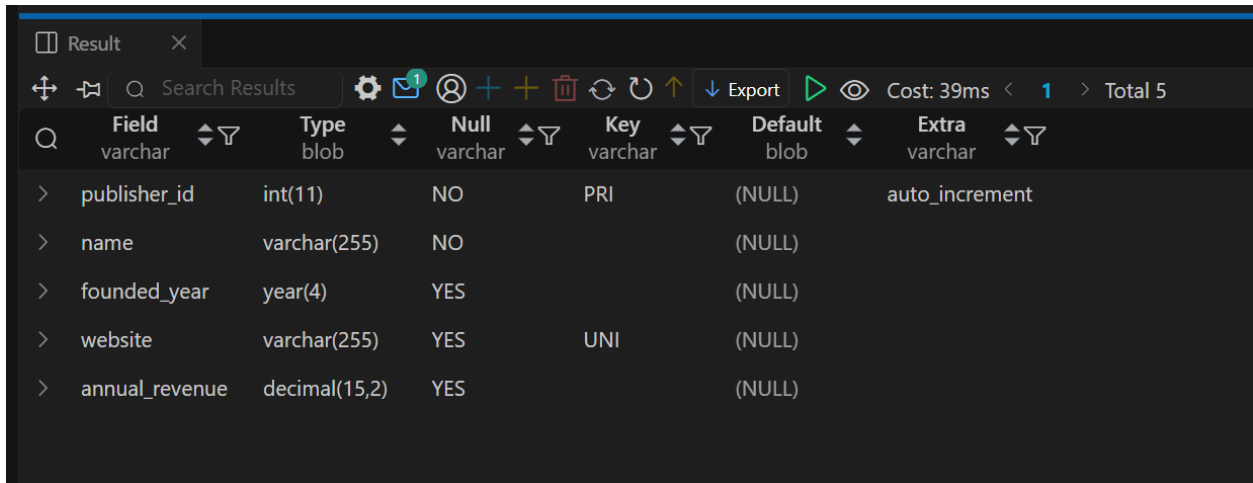
Operation 05

```
CREATE TABLE books (  
  book_id INT NOT NULL AUTO_INCREMENT,  
  title VARCHAR(255) NOT NULL,  
  isbn VARCHAR(20) UNIQUE,  
  price DECIMAL(8,2),  
  publication_date DATE,  
  page_count INT,  
  publisher_id INT,  
  PRIMARY KEY (book_id),  
  CONSTRAINT fk_books_publisher  
  FOREIGN KEY (publisher_id)  
  REFERENCES publishers(publisher_id)  
);
```

! 17 CREATE TABLE books (Table 'books' already exists
18 book_id INT NOT NULL AUTO_INCREMENT,
19 title VARCHAR(255) NOT NULL,
20 isbn VARCHAR(20) UNIQUE,
21 price DECIMAL(8,2),
22 publication_date DATE,
23 page_count INT,
24 publisher_id INT,
25 PRIMARY KEY (book_id),
26 CONSTRAINT fk_books_publisher
27 FOREIGN KEY (publisher_id)
28 REFERENCES publishers(publisher_id)
29); 3ms
30

Operation 06

DESCRIBE publishers;



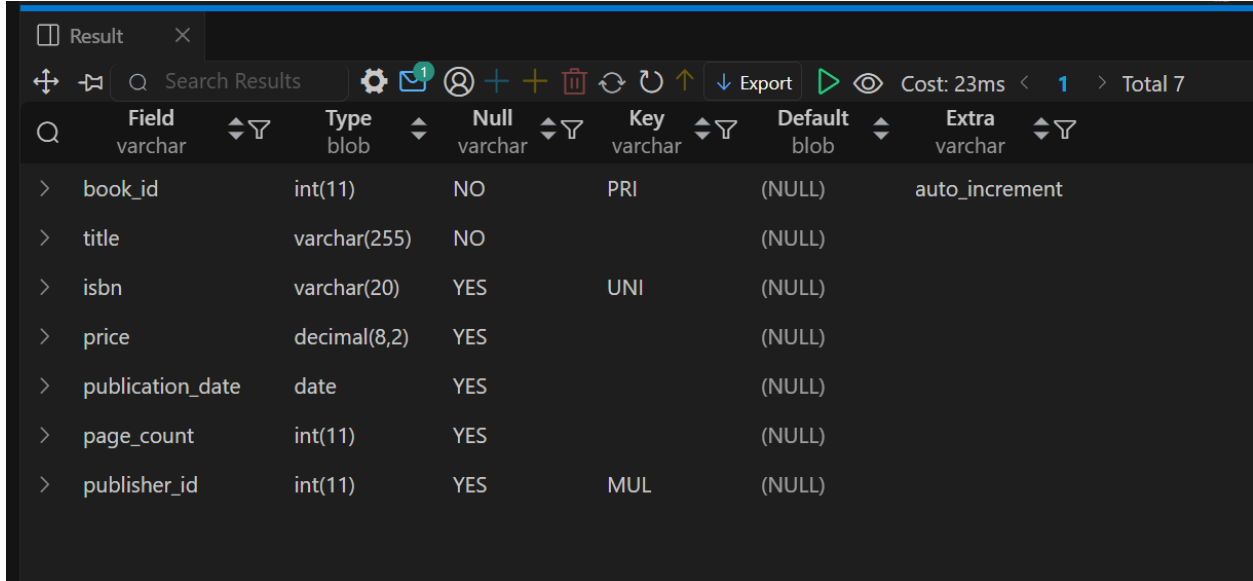
Result

Search Results

Cost: 39ms < 1 > Total 5

Field	Type	Null	Key	Default	Extra
varchar	blob	varchar	varchar	blob	varchar
> publisher_id	int(11)	NO	PRI	(NULL)	auto_increment
> name	varchar(255)	NO		(NULL)	
> founded_year	year(4)	YES		(NULL)	
> website	varchar(255)	YES	UNI	(NULL)	
> annual_revenue	decimal(15,2)	YES		(NULL)	

DESCRIBE books;



Result

Search Results

Cost: 23ms < 1 > Total 7

Field	Type	Null	Key	Default	Extra
varchar	blob	varchar	varchar	blob	varchar
> book_id	int(11)	NO	PRI	(NULL)	auto_increment
> title	varchar(255)	NO		(NULL)	
> isbn	varchar(20)	YES	UNI	(NULL)	
> price	decimal(8,2)	YES		(NULL)	
> publication_date	date	YES		(NULL)	
> page_count	int(11)	YES		(NULL)	
> publisher_id	int(11)	YES	MUL	(NULL)	

Operation 07

```
ALTER TABLE books  
ADD COLUMN genre VARCHAR(100);
```

▷Run

✓ 37 ALTER TABLE books

38 ADD COLUMN genre VARCHAR(100); 6ms

39

Operation 08

```
ALTER TABLE books  
ADD CONSTRAINT chk_page_count_positive  
CHECK (page_count > 0);
```

▷Run | ☐Select

✓ 41 ALTER TABLE books

42 ADD CONSTRAINT chk_page_count_positive

43 CHECK (page_count > 0); 54ms

44

Operation 09

```
ALTER TABLE publishers  
ADD CONSTRAINT chk_founded_year  
CHECK (founded_year >= 1800);
```

▷Run | ☐Select

✓ 46 ALTER TABLE publishers

47 ADD CONSTRAINT chk_founded_year

48 CHECK (founded_year >= 1800); 48ms

49

▷Run

Operation 10

DESCRIBE publishers;

Field	Type	Null	Key	Default	Extra
publisher_id	int(11)	NO	PRI	(NULL)	auto_increment
name	varchar(255)	NO		(NULL)	
founded_year	year(4)	YES		(NULL)	
website	varchar(255)	YES	UNI	(NULL)	
annual_revenue	decimal(15,2)	YES		(NULL)	

DESCRIBE books;

Field	Type	Null	Key	Default	Extra
book_id	int(11)	NO	PRI	(NULL)	auto_increment
title	varchar(255)	NO		(NULL)	
isbn	varchar(20)	YES	UNI	(NULL)	
price	decimal(8,2)	YES		(NULL)	
publication_date	date	YES		(NULL)	
page_count	int(11)	YES		(NULL)	
publisher_id	int(11)	YES	MUL	(NULL)	
genre	varchar(100)	YES		(NULL)	